

Embodied AI: Designing Inclusive Digital Ecologies

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Abstract. Contemporary approaches to Artificial Intelligence in Education (AIED) are predominantly grounded in representational models that emphasize prediction, personalization, and learning optimization. This paper proposes an alternative perspective by integrating Embodied Interaction, Structural Coupling, Ecological Dynamics, Embodied Design, and the Special Education Embodied Design (SpEED) framework. We introduce Embodied AI as an ecological approach to Human–Computer Interaction in which AI is understood not as a source of instruction or knowledge delivery, but as a participant in embodied systems of practice. Within this perspective, meaning emerges through sociomaterial coordination among learners, technologies, environments, and cultural practices. The framework is illustrated through two inclusive mathematics learning environments, QUAD and Graph-Math, originally designed for blind learners. Rather than treating accessibility as accommodation, these cases demonstrate how inclusive design can function as a source of epistemic innovation, expanding opportunities for participation and revealing alternative pathways for mathematical meaning-making. We argue that the future challenge of educational AI is not prediction but participation, and that Embodied AI provides a foundation for designing inclusive ecological systems of Human–Computer Interaction in which intelligence emerges through evolving relations among bodies, technologies, and environments.

1 INTRODUCTION

Artificial intelligence is rapidly reshaping contemporary educational systems. Intelligent tutoring systems, adaptive learning platforms, learning analytics, and generative AI applications are typically grounded in the assumption that learning can be enhanced through increasingly accurate models of learner behavior and the efficient delivery of instructional support (OECD, 2025). While these approaches have produced powerful technological innovations, they remain largely informed by representational and information-processing theories of cognition, in which intelligence is understood as the manipulation of internal models and learning as the acquisition of abstract knowledge structures. In contrast, contemporary developments in embodied cognition, ecological psychology, enactivism, and Human–Computer Interaction suggest that cognition does not reside exclusively within individuals or computational systems, but emerges through ongoing interactions among bodies, environments, artifacts, and sociocultural practices (Varela, Thompson, & Rosch, 1991; Thompson, 2007; Gibson, 1979; Chemero, 2009). From this perspective, meaning arises not through the passive reception of

information but through active participation in systems of perception and action.

These ideas have been further developed through theories of structural coupling, ecological dynamics, and embodied learning. Structural coupling proposes that cognition emerges through recurrent histories of organism–environment interaction, where learning is understood as the transformation of relational dynamics rather than the internalization of representations (Maturana & Varela, 1987; Varela, 1997). Ecological dynamics similarly conceptualizes learning as an emergent process arising from interactions among agent, task, and environmental constraints (Araújo & Davids, 2011; Newell & Ranganathan, 2010). Within mathematics education, these commitments have inspired Embodied Design, a pedagogical framework that investigates how mathematical concepts emerge through coordinated interactions among bodies, artifacts, movements, and symbolic practices (Abrahamson & Sánchez-García, 2016). More recently, the Special Education Embodied Design (SpEED) framework has extended these insights into inclusive educational contexts by foregrounding the embodied resources of learners whose sensory, cognitive, or communicative experiences are often marginalized within conventional learning environments (Tancredi et al., 2021, 2022; Abrahamson et al., 2023).

Despite these advances, their implications for artificial intelligence and Human–Computer Interaction remain underdeveloped (Videla, Penny & Ross, 2025). Educational AI continues to be predominantly framed in terms of prediction, personalization, and automated instruction, while HCI often focuses on usability, efficiency, or information exchange. Such perspectives overlook the possibility that intelligent technologies may participate directly in the emergence of meaning through embodied systems of practice. This paper addresses this gap by proposing the concept of Embodied AI. Drawing upon Embodied Interaction (Dourish, 2001), Structural Coupling, Ecological Dynamics, Embodied Design, and SpEED, we conceptualize AI not as a source of knowledge delivery but as an ecological participant within sociomaterial learning systems. From this perspective, AI contributes to the ongoing reorganization of affordances, constraints, modalities, and opportunities for action through which meaning emerges. We illustrate the explanatory potential of this framework through two inclusive mathematics learning environments, QUAD and Graph-Math, and argue that the future challenge of educational AI is not prediction but participation: designing intelligent technologies that support embodied, inclusive, and meaningful forms of Human–Computer Interaction.

2 FROM REPRESENTACION TO PARTICIPATION

Contemporary developments in Artificial Intelligence for Education (AIED) have been largely driven by advances in machine learning, predictive analytics, adaptive systems, and generative models. Intelligent tutoring systems, learner models, recommendation engines, and generative AI applications are typically designed around the capacity to infer internal states and deliver personalized instructional interventions. Within this paradigm, cognition is often understood as the processing and manipulation of information, while learning is conceived as the acquisition or refinement of internal representations.

Over the past several decades, a growing body of research has challenged the adequacy of representational accounts for understanding human cognition and learning (Gallagher, 2017). Perspectives emerging from embodied cognition, ecological psychology, enactivism, and phenomenology have argued that cognition cannot be reduced to internal information processing (Hutto y Myin, 2013). A foundational contribution to this shift was provided by ecological psychology. Gibson (1979) rejected the notion that perception requires the internal reconstruction of an external world through mental representations. Instead, he proposed that organisms directly perceive affordances, opportunities for action available within specific environmental conditions (Turvey, 1992). Perception is therefore inseparable from action. Organisms do not passively receive information and subsequently interpret it, they actively explore environments in ways that reveal possibilities for engagement (Malafouris, 2013). Meaning emerges not from internal representations but from practical relations between organisms and their surroundings.

Similar critiques emerged within embodied and enactive approaches to cognition. Varela, Thompson, and Rosch (1991) argued that cognition should be understood as embodied action rather than symbolic computation. From an enactive perspective, organisms do not represent a pre-given world, they enact meaningful worlds through histories of structural coupling with their environments (Maturana y Varela, 1980; Hutto y Myin, 2017). These developments have profound implications for Human–Computer Interaction. If cognition is fundamentally embodied, situated, and relational, then interaction cannot be understood merely as information exchange between users and computational systems. Instead, interaction becomes a process through which meaning emerges in practice (Dourish, 2001). This shift is particularly significant in educational contexts, where learning technologies are often evaluated according to their capacity to deliver information, provide feedback, or improve measurable outcomes. Such criteria may obscure more fundamental questions concerning how technologies participate in the constitution of activity itself.

Learners do not encounter educational environments as isolated cognitive agents. They participate in complex ecologies involving bodies, artifacts, representations, cultural practices, and other people. Understanding these dynamics requires moving beyond representational accounts of cognition toward a participation-centered understanding of interaction. In this paper, we adopt participation rather than representation as the primary unit of analysis. Participation refers to the ways in which

learners become engaged in embodied systems of practice through ongoing interactions with people, technologies, materials, and environments. From this perspective, the central challenge for educational technology is not how to improve the transmission of information, but how to support meaningful forms of participation. This shift provides the conceptual foundation for rethinking artificial intelligence as a component of embodied learning ecologies rather than as an external source of instruction or knowledge. This reconceptualization has important implications for understanding interaction, learning, and technology. If cognition is fundamentally embodied and relational, interaction cannot be reduced to the transmission of information between independent entities. Human activity unfolds within ecological systems composed of bodies, artifacts, representations, technologies, cultural practices, and other people, all of which participate in shaping possibilities for action.

3 EMBODIED INTERACTION AND HUMAN-COMPUTER INTERACTION

The shift from representation to participation requires a corresponding reconceptualization of Human–Computer Interaction. If cognition is understood as emerging through embodied engagement with the world, then interaction cannot be reduced to the exchange of information between users and computational systems. Instead, interaction becomes a process through which meaning is enacted, negotiated, and transformed within ongoing systems of practice. A particularly influential contribution to this perspective was provided by Dourish (2001), whose theory of embodied interaction challenged dominant computational models of HCI by grounding interaction in phenomenological accounts of human experience.

Drawing upon Heidegger, Merleau-Ponty, Schutz, and Wittgenstein, Dourish argued that interaction should not be understood primarily in terms of representations, commands, or information transfer. Rather, interaction emerges through embodied participation in meaningful activities. Technologies become intelligible not because users possess internal models of how they operate, but because they become incorporated into practical engagements with the world. Meaning is not located inside computational representations or mental structures; it emerges through situated action.

Central to Dourish's account is the concept of coupling. Human beings do not simply use tools; they become coupled with them through practice. As tools become integrated into activity, they cease to function as external objects and instead become components of ongoing action. Dourish illustrates this point through the example of writing with a pen. Skilled writers do not consciously coordinate individual finger movements or monitor the pen as an object, rather, they become engaged in the activity of writing itself. Interaction, therefore, is not a relationship between separate entities but a process of mutual coordination among bodies, artifacts, and environments.

This understanding challenges persistent assumptions within contemporary digital education and artificial intelligence. Many educational technologies continue to be designed as systems that deliver information, provide feedback, or optimize user

performance. Such approaches implicitly position technology as an external mediator between knowledge and learner. Dourish's (2001) perspective argues that meaning is not imposed by designers, encoded within interfaces, or determined by computational representations, therefore, interaction cannot be fully specified in advance. Technologies create possibilities for action, but the significance of those possibilities emerges through participation in practice. This perspective resonates strongly with ecological approaches to perception and action, where affordances are understood not as fixed properties of environments but as relational opportunities that emerge through organism–environment engagement.

The educational significance of a technology cannot be understood solely in terms of its informational content or instructional effectiveness. It must also be understood in terms of how it reorganizes opportunities for action, perception, exploration, and participation. Embodied interaction also challenges the notion of context as an external container surrounding activity. For Dourish (2001), context and activity are mutually constitutive. This insight has important implications for educational technology design, since learning environments cannot be understood as neutral settings within which learning occurs. Instead, they are dynamically enacted through ongoing interactions among learners, teachers, technologies, materials, and institutional practices.

Taken together, these ideas suggest a fundamentally different understanding of Human–Computer Interaction. Interaction is not information exchange. It is embodied participation in systems of practice. Technologies are not neutral tools that mediate between users and information. Rather than conceptualizing AI as an autonomous source of knowledge, prediction, or instruction, we propose that artificial intelligence should be understood as part of the sociomaterial ecology through which interaction and meaning emerge. This proposal provides the foundation for integrating embodied interaction with ecological dynamics, structural coupling, Embodied Design, and inclusive educational frameworks.

4 STRUCTURAL COUPLING AND RELATIONAL DYNAMICS OF INTERACTION

A complementary account of interaction is provided by the Biology of Cognition developed by Maturana and Varela (1980, 1987). This framework begins with a radical departure from representational theories of mind by grounding cognition in the organizational properties of living systems. A central implication of this view is that living systems are structurally determined systems. At any given moment, all changes that occur within an organism are determined by its current structure rather than by external inputs. Environmental events do not instruct organisms, transmit information into them, or determine their internal states. Rather, environmental perturbations trigger structural changes whose specific consequences depend on the organism's own organization and structural history. Consequently, cognition cannot be understood as the reception or processing of information from an objective external world. What an organism

does in response to a perturbation is always determined by its own structure at that moment.

Within this framework, structural coupling describes the historical process through which organism and environment undergo mutually congruent transformations through recurrent interactions (Maturana y Varela, 1980). Through continued engagement, organisms become increasingly attuned to the regularities of their environments, while environments simultaneously become structured through the activities of organisms. Over time, these reciprocal adjustments generate domains of viable interaction in which organism and environment become dynamically coordinated. Structural coupling is therefore not a connection between pre-existing entities but an ongoing process through which relational coherence emerges and is maintained.

From this perspective, cognition is not the representation of an external reality but the enactment of a meaningful world through histories of interaction. Organisms do not discover a pre-given world; they bring forth domains of significance through recurrent cycles of perception and action. Meaning emerges within the relational dynamics of coupling itself, cognition is therefore fundamentally embodied, situated, and historical. What an organism can perceive, discriminate, and coordinate depends on the history of couplings through which its current structure has been constituted. These ideas have been further developed within contemporary enactivist and ecological approaches to cognition. Radical and extensive enactivism argue that cognition should not be understood as the manipulation of internal representational content but as the unfolding of embodied organism–environment dynamics (Hutto & Myin, 2013; Hutto et al., 2014). Cognitive processes are not internal events occasionally extended into the world; rather, they are constitutively relational processes distributed across evolving systems of interaction. This shift further reinforces the view that cognition is enacted through participation rather than generated inside individual minds.

Recent developments in Embodied AI have extended this framework by distinguishing among constitutive, enabling, and contextual factors involved in cognitive phenomena (Videla, Penny and Ross, 2025). Constitutive factors are those processes that directly participate in the production of a phenomenon. Enabling factors create the conditions that make the phenomenon possible, while contextual factors modulate its expression without directly producing it. Learning, from this perspective, emerges through dynamic interactions among these domains rather than from isolated causal mechanisms. This understanding has significant implications for Human–Computer Interaction. Technologies should not be viewed as external devices that transmit information into learners. Nor should artificial intelligence be understood as a system that directly causes learning. Instead, technological systems participate as components within broader ecologies of structural coupling. They introduce perturbations, reorganize constraints, and modify the affordances available within ongoing activity. Their influence is always mediated by the structural dynamics of the systems in which they participate.

Consequently, the educational significance of artificial intelligence lies not in its capacity to represent knowledge more accurately or provide more efficient feedback. Rather, its significance lies in its capacity to participate in evolving systems of structural coupling that support new forms of embodied interaction. AI becomes part of the relational dynamics through which learners, teachers, artifacts, and environments continuously co-constitute one another. Understanding these dynamics requires moving beyond representational accounts of technology toward an ecological conception of interaction grounded in participation, coupling, and the emergence of meaning.

5 EMBODIED DESIGN AND INCLUSIVE LEARNING ECOLOGIES

Embodied Design emerged as a response to the limitations of representationalist theories of mathematical learning. Rather than conceiving learning as the progressive acquisition of abstract representations, this framework proposes that mathematical understanding arises through the reorganization of perception–action systems within carefully structured environments (Abrahamson, 2009, 2014). Drawing on embodied cognition, ecological psychology, enactivism, and dynamical systems theory, Embodied Design conceptualizes learning as a relational, situated, and embodied process grounded in interactions among learners, artifacts, environments, and cultural practices.

Within this perspective, mathematical concepts emerge through the stabilization of perceptual and motor coordinations that become meaningful within activity (Abrahamson & Sánchez-García, 2016). Mathematical knowing is therefore not primarily a matter of acquiring symbolic content but of developing new ways of perceiving, acting, discriminating, and coordinating within mathematically organized environments. This view aligns closely with extensive enactivism, which rejects the assumption that cognition is fundamentally representational and instead understands cognition as world-involving from the outset (Hutto & Myin, 2013; Hutto, 2017).

Rather than treating mathematical ideas as external abstractions, Embodied Design creates conditions in which learners can experience conceptual relations through embodied activity. Learning emerges through a gradual alignment between evolving sensorimotor coordinations and culturally valued forms of mathematical reasoning. Embodied Design operationalizes these commitments through the deliberate configuration of learning ecologies. Drawing on ecological dynamics, learning is understood as emerging from interactions among agent constraints, environmental constraints, and task constraints (Newell, 1986; Newell & Ranganathan, 2010). Agent constraints include learners' embodied histories and perceptual capacities; environmental constraints involve material, digital, spatial, and social conditions; and task constraints refer to goals, rules, and forms of feedback. Learning is not produced through information transmission but through adaptive reorganization within these interacting constraints.

Within this framework, perception is understood as attunement to affordances (Gibson, 1979). Educational design therefore does not impose meaning but structures environments in which

meaningful affordances can emerge. Designers create opportunities for learners to encounter and stabilize mathematically relevant relations through embodied engagement. In this sense, Embodied Design can be understood as the pedagogical operationalization of ecological dynamics, translating principles of perception–action coupling into educational practice (Abrahamson et al., 2023).

A central contribution of this framework is the concept of attentional anchors (Abrahamson et al., 2011; Abrahamson & Sánchez-García, 2016). Attentional anchors are perceptually salient and action-relevant structures that learners spontaneously develop in order to coordinate activity within mathematically organized tasks. Rather than functioning as symbolic representations, they serve as ecological control structures that stabilize perception–action loops and support the detection of mathematically relevant invariants. Through interaction with materials, technologies, and social partners, attentional anchors become progressively refined, enabling increasingly sophisticated forms of coordination that later ground formal mathematical reasoning (Abrahamson & Bakker, 2016; Abrahamson et al., 2020).

This perspective has important implications for educational technology. Technologies are not merely vehicles for delivering information or representing concepts. Instead, they participate directly in the organization of perception–action systems and become components of evolving ecologies of structural coupling. Their educational significance lies not in the content they communicate but in the forms of participation they make possible. From this standpoint, artificial intelligence can be reconceptualized as a dynamic modulator of ecological constraints. Rather than functioning as a source of explanations or instructional authority, AI can reorganize affordances, amplify perceptual cues, support multimodal engagement, and sustain emerging attentional anchors without eliminating exploratory activity. AI thus becomes part of a distributed cognitive ecology that supports conceptual emergence while preserving learner agency. Its role is not to replace human reasoning but to participate in the ongoing reconfiguration of conditions under which meaningful action becomes possible.

The relevance of this perspective becomes particularly evident in inclusive educational contexts. Conventional educational environments often privilege narrow sensory, motor, and representational pathways, marginalizing learners whose embodied experiences diverge from normative assumptions. Embodied Design challenges this orientation by treating bodily diversity not as a limitation but as a resource for design. Different sensory and cognitive profiles reveal alternative affordances, alternative coordinations, and alternative pathways toward conceptual understanding. In this sense, inclusion is not merely a matter of accessibility but a source of epistemic innovation. Research with blind and visually impaired learners has demonstrated that accessibility can generate new theoretical insights into perception, coordination, and learning (Abrahamson et al., 2019). More recently, Abrahamson et al. (2023) has described this orientation as Inclusive Design for Expansive Theory, arguing that educational researchers can develop more powerful theories of learning by designing from the embodied resources of diverse learners. Rather than adapting existing

models to accommodate difference, this perspective treats diversity itself as a generative condition for theoretical and pedagogical innovation. Viewed through this lens, Embodied Design is not merely a pedagogical methodology for mathematics education. It is an ecological design framework for intentionally configuring sociomaterial conditions under which new forms of perception, participation, structural coupling, and meaning-making can emerge. This orientation provides the conceptual bridge toward the Special Education Embodied Design (SpEED) framework and, ultimately, toward our conception of Embodied AI as an ecological approach to inclusive Human–Computer Interaction.

6 THE SPEED FRAMEWORK: TOWARD AN ECOLOGICAL MODEL OF INCLUSIVE HUMAN-COMPUTER INTERACTION

While Embodied Design provides a powerful framework for understanding how mathematical meanings emerge through embodied participation, it offers fewer explicit resources for addressing the diversity of sensory, motor, and cognitive experiences present in contemporary educational environments. This challenge becomes particularly important when designing technologies intended to support learners whose perceptual and interactional profiles diverge from normative assumptions. In response to this challenge, researchers have developed the Special Education Embodied Design (SpEED) framework, an extension of Embodied Design that places embodied diversity at the center of educational innovation (Tancredi et al., 2021, 2022).

SpEED emerged from a growing recognition that conventional approaches to inclusion often treat accessibility as a process of adapting existing learning environments to accommodate individual differences. In contrast, SpEED begins from the assumption that perceptual diversity constitutes a productive resource for educational design. Learners who are blind, autistic, neurodivergent, or otherwise marginalized frequently reveal dimensions of interaction that remain invisible within standard educational settings. Rather than viewing these learners through a deficit lens, SpEED treats their embodied experiences as generative starting points for the design of new learning ecologies. Abrahamson et al. (2019) demonstrated how tactile mathematical activities involving blind learners challenge traditional assumptions about mathematical perception and reveal alternative pathways toward conceptual understanding.

Subsequent work argued that accessibility should not be understood merely as accommodation but as a source of theoretical and pedagogical innovation (Tancredi et al., 2021, 2022; Abrahamson, 2023). This shift reframes inclusion as a process of expanding the ecological possibilities for participation rather than adapting learners to existing instructional structures. SpEED operationalizes this perspective through three interrelated design parameters: media, modalities, and semiotic modes. Together, these dimensions provide a framework for analyzing and designing embodied learning environments while also offering a powerful model for understanding Human–Computer Interaction within inclusive educational ecologies.

From Tancredi et al (2022) the first parameter, media, refers to the material and technological resources that participate in activity. Media include physical artifacts, digital interfaces, tangible manipulatives, sonification systems, sensors, intelligent technologies, and other components that become integrated into embodied systems of practice. Importantly, media are not understood as passive instructional supports. Consistent with Dourish’s (2001) conception of embodied interaction, they participate directly in the constitution of activity itself. Media shape what can be perceived, coordinated, explored, and enacted within a learning ecology.

The second parameter, modalities, refers to the sensory and motor channels through which learners engage with the environment. Vision, touch, audition, proprioception, kinesthetic awareness, gesture, and movement all constitute modalities through which perception and action become organized. SpEED recognizes that learners do not access educational environments through identical sensory pathways. Different modalities reveal different affordances and support different forms of participation. Consequently, inclusive design requires attention not only to what learners know but also to how they perceive, act, and coordinate within educational activity.

The third parameter, semiotic modes, concerns the representational and communicative forms through which meaning is expressed and negotiated (Vygotsky, 1962). These include gestures, embodied actions, diagrams, spoken language, tactile representations, symbolic notations, and digital visualizations. Within SpEED, semiotic modes are not viewed as endpoints of cognition but as resources that emerge from and remain grounded in embodied activity. Mathematical symbols, for example, are understood as culturally stabilized forms of coordination that become meaningful through their relation to embodied action and perception (Abrahamson, 2021). From the perspective developed in this paper, SpEED can be understood as an ecological model of Human–Computer Interaction. Its significance extends beyond special education because it provides a concrete vocabulary for analyzing how technologies participate in embodied systems of practice. This ecological interpretation of SpEED also provides a bridge between inclusive design and Human–Computer Interaction. Inclusion is no longer understood as a specialized concern applicable only to particular learner populations. Instead, it becomes a methodological strategy for expanding the range of affordances available within sociomaterial systems. By designing from the embodied resources of diverse learners, researchers gain access to richer forms of interaction, more flexible configurations of participation, and ultimately more expansive theories of cognition and learning.

7 EMBODIED AI: AN ECOLOGICAL FRAMEWORK FOR INCLUSIVE HUMAN-COMPUTER INTERACTION

The theoretical perspectives reviewed thus far converge on a common proposition: cognition does not reside exclusively within individual minds or computational systems but emerges through embodied participation in sociomaterial ecologies. From perspectives Embodied Interaction (Dourish, 2001), Ecological

Dynamics (Araújo & Davids, 2011), Structural Coupling (Maturana & Varela, 1987). Embodied Design (Abrahamson, 2014) and SpEED (Tancredi et al., 2022)., we propose Embodied AI as an ecological framework for Human–Computer Interaction. Rather than conceptualizing artificial intelligence as a source of instruction, prediction, personalization, or knowledge delivery, Embodied AI understands AI systems as participants within embodied systems of practice. Their role is not to replace human cognition, optimize learning, or automate decision making. Instead, intelligent systems participate in the ongoing reorganization of affordances, constraints, modalities, and semiotic resources through which meaning emerges. Consequently, the central question shifts from “What knowledge does AI provide?” to “How does AI participate in the organization of meaningful activity?”

This framework becomes particularly relevant within inclusive educational contexts, where diversity in sensory, motor, and cognitive experience reveals alternative pathways for participation and meaning-making. To illustrate the explanatory potential of Embodied AI, we examine two embodied mathematics learning environments: QUAD and Graph-Math. These systems were originally developed to support blind learners, yet their significance extends beyond accessibility. They demonstrate how designing from marginalized embodied experiences can generate more expansive forms of Human–Computer Interaction for all learners.

8 ILLUSTRATIVE CASES OF EMBODIED AI

QUAD: Reconfiguring Geometric Interaction Through Embodied AI



Figure 1: QUAD Case

QUAD was initially conceived as an embodied mathematics environment that enables blind and low-vision learners to explore geometric invariants through tactile and proprioceptive interaction. Unlike conventional approaches that translate visual diagrams into verbal descriptions, QUAD reorganizes the learning ecology itself. Geometry is no longer treated as a visual object to be represented but as a set of relational structures that can be enacted through embodied exploration. The system consists of a sensorized quadrilateral whose sides and vertices can be manipulated physically. These movements are continuously processed through a mechatronic architecture that

couples tactile action with computational interpretation and auditory feedback. Within a traditional Human–Computer Interaction framework, the device might be understood as an interface for transmitting geometric information. From an Embodied AI perspective, however, QUAD functions as a sociomaterial ecology in which learners, artifacts, computational processes, and mathematical relations become structurally coupled through activity.

Analyzed through SpEED, QUAD reveals how embodied participation emerges through the interaction of multiple design parameters. As media, the manipulable artifact, sensors, computational infrastructure, and auditory feedback system operate as active components of the cognitive ecology. As modalities, tactile, proprioceptive, kinesthetic, and auditory channels become primary resources for perception and coordination. As semiotic modes, geometric categories such as square, rectangle, trapezoid, or parallelogram emerge only after learners stabilize embodied relations among movement, shape transformation, and auditory cues. Within this ecology, AI does not function as an evaluator of correctness. Rather, it participates in the stabilization of interaction. By identifying relational transformations and maintaining coherent correspondences between tactile manipulations and auditory responses, the intelligent system helps sustain the perception–action dynamics through which geometric invariants become available for exploration. The significance of AI lies not in providing answers but in supporting the ecological conditions that allow learners to discover and enact mathematical relations.

Importantly, QUAD illustrates a broader principle of expansive inclusive design. What began as a design for blind learners reveals fundamental aspects of geometric cognition that conventional visual instruction often obscures. The system demonstrates that geometric concepts are not fundamentally visual representations but embodied invariants emerging through interaction. Designing from the perceptual resources of blind learners therefore expands the theoretical understanding of geometry itself, generating richer possibilities for participation across diverse learner populations.

Graph-Math: Embodied Functional Reasoning Through Multimodal Coordination



Figure 2: Graph Math Case

Graph-Math extends these principles into the domain of analytic geometry and functional reasoning. Originally developed to support blind students in exploring Cartesian relations without

reliance on visual graphs, the system establishes a dynamic correspondence between bodily action, tactile exploration, computational processing, and mathematical conceptualization. At the center of the environment is a 13×13 sensorized tactile matrix composed of 169 pressure-sensitive nodes connected to an Arduino-based computational architecture. Learners physically position markers within the grid, generating real-time mappings onto a digital Cartesian plane. Through repeated interaction, coordinate relations, slopes, intercepts, and functional transformations become available as embodied experiences rather than abstract visual objects.

Within the Embodied AI framework, Graph-Math is not a tool for translating visual graphs into alternative sensory formats. Instead, it constitutes an ecological environment in which mathematical meanings emerge through the coordination of action, perception, and representation. The learner contributes embodied histories, exploratory strategies, and sensorimotor capacities. The task involves investigating relationships among coordinates, rates of change, and functional trajectories. The environment consists of a tightly coupled network of tactile surfaces, computational processing, auditory feedback, and graphical representation.

The SpEED framework illuminates these dynamics. As media, Graph-Math integrates tactile hardware, computational infrastructure, sonification systems, and graphical displays. As modalities, tactile, kinesthetic, auditory, and visual channels become coordinated within a shared perception-action system. As semiotic modes, symbolic notation, coordinate pairs, graphs, gestures, and spoken language emerge as interconnected resources for mathematical reasoning. The role of AI within Graph-Math differs significantly from that of conventional intelligent tutoring systems. Rather than diagnosing misconceptions or delivering corrective instruction, the AI maintains structural correspondences across modalities and representations. When learners position markers inconsistently with the targeted mathematical relations, the system generates exploratory prompts that invite further investigation rather than prescribing solutions. AI thus functions as an ecological participant that helps sustain coherent relations among embodied actions, tactile configurations, auditory cues, and graphical representations.

Through repeated interaction, learners develop attentional anchors that enable them to perceive slope as inclination, intercept as boundary relation, and functional change as a dynamic transformation of trajectories. Mathematical meaning emerges not from the interpretation of representations but from embodied participation in a multimodal ecology of action. As in QUAD, the significance of the design extends beyond accessibility. By foregrounding tactile and kinesthetic interaction, Graph-Math reveals alternative epistemic pathways that enrich mathematical learning for all learners. Embodied AI as Inclusive Human-Computer Interaction Together, QUAD and Graph-Math illustrate a shift from instructional AI toward ecological AI. In both cases, artificial intelligence does not serve as an evaluator, or predictive system. Instead, it participates in embodied ecologies of interaction, supporting the emergence of affordances, stabilizing multimodal coordinations, and sustaining opportunities for participation.

7 CONCLUSIONS

This paper has proposed Embodied AI as an ecological framework for inclusive Human-Computer Interaction. Drawing upon Embodied Interaction, Structural Coupling, Ecological Dynamics, Embodied Design, and the SpEED framework, we have argued that artificial intelligence should not be understood primarily as a mechanism for prediction, personalization, or automated instruction. Instead, AI can be conceived as a participant within embodied systems of practice that contribute to the ongoing reorganization of affordances, constraints, modalities, and opportunities for action through which meaning emerges. The cases of QUAD and Graph-Math illustrate how intelligent technologies can support multimodal participation and embodied forms of mathematical reasoning while preserving learner agency. More broadly, they demonstrate that designing from the embodied resources of marginalized learners does not merely increase accessibility; it expands the theoretical understanding of cognition, interaction, and learning itself. Ultimately, the contribution of Embodied AI lies in shifting the focus of educational technology from optimization to participation, and from information delivery to the creation of inclusive sociomaterial ecologies in which learners, technologies, and environments jointly enact meaningful worlds.

7 DISCUSSIONS: WHAT MAKES AI EMBODIED?

A central question emerging from this work concerns why the proposed framework should be described as Embodied AI. After all, both QUAD and Graph-Math could function without sophisticated artificial intelligence. Their educational value does not depend on automated instruction, predictive analytics, or learner modeling. This observation reveals an important conceptual distinction. In conventional educational technology, AI is typically defined by its capacity to classify, predict, recommend, or personalize. Within the Embodied AI framework, however, AI is defined by the role it plays within embodied systems of practice.

Artificial intelligence becomes embodied not when it simulates human cognition, but when it participates in the organization of perception-action systems. More specifically, AI becomes embodied when it sustains multimodal correspondences across tactile, auditory, visual, and kinesthetic channels; supports the emergence and stabilization of attentional anchors; modulates ecological constraints without eliminating exploratory activity; reorganizes affordances that shape possibilities for participation; and participates in evolving histories of structural coupling among learners, artifacts, and environments. In this sense, AI is not conceived as an external tutor or intelligent authority but as an ecological participant within sociomaterial systems of interaction.

This perspective extends Dourish's notion of embodied interaction beyond traditional Human-Computer Interaction by incorporating contemporary developments in ecological dynamics, enactivism, and inclusive design. Rather than evaluating AI according to the accuracy of its predictions or the efficiency of its instructional interventions, Embodied AI redirects attention toward the ways intelligent technologies participate in the emergence of meaningful activity. The significance of AI therefore lies not in what it knows, but in how

it contributes to the ongoing organization of participation, coordination, and sense-making. Such a shift opens new directions for Human–Computer Interaction research, particularly within collaborative, creative, and inclusive learning environments where meaning emerges through distributed systems of embodied action rather than through information transmission alone.

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